Cte_Join

30 January 2026

I Have used Ubuntu for running Gurivi_Reddy_Yasaswini_Cte_and_Join.sql and Gurivi_Reddy_Yasaswini_Cte_and_Join_output.txt Is ouptu file

```
➜ cte and join ls
 Dataset_given_sql_databasesql-d5pqoasrtl0hpva4650g.sql  'Gurivi_Reddy_Yasaswini_Cte and Join.sql'   cteandjoinsmd-d5tg704rtl0r3n14chm0.md
➜ cte and join sudo mysql

[sudo] password for yasaswini:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.44-0ubuntu0.22.04.2 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE cte_join_db;
b;
Query OK, 1 row affected (0.02 sec)

mysql> USE cte_join_db;
Database changed
mysql>
```

You have Docker installed on your system. Do
install the recommended 'Dev Containers' ext
Microsoft for it?

Install

The terminal process failed to launch: Starting
"C:\Users\gyasu\Desktop\Calibo\sql
basic\GuriviReddyYasaswini_JOIN, GROUP
BY\GuriviReddyYasaswini_JOIN, GROUP BY\Al

Loaded the data set given by INSTRUCTOR

sql basic

cteandjoinsmd-d5tg704rtl0r3n14chm0.md | Dataset_given_sql_databasesql-d5pqoasrtl0hpva4650g.sql | Gurivi_

cte and join > Dataset_given_sql_databasesql-d5pqoasrtl0hpva4650g.sql

```
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-- California, 94105, USA.
--
--  DISCLAIMER
--  To the best of our knowledge, this data is fabricated, and
--  it does not correspond to real people.
--  Any similarity to existing people is purely coincidental.
--

DROP DATABASE IF EXISTS employees;
CREATE DATABASE IF NOT EXISTS employees;
USE employees;

SELECT 'CREATING DATABASE STRUCTURE' as 'INFO';

DROP TABLE IF EXISTS dept_emp,
                     dept_manager,
                     titles,
                     salaries,
                     employees,
                     departments;

/*!50503 set default_storage_engine = InnoDB */;
/*!50503 select CONCAT('storage engine: ', @@default_storage_engine) as INFO */;

CREATE TABLE employees (
    emp_no      INT             NOT NULL,
    birth_date  DATE            NOT NULL,
    first_name  VARCHAR(14)     NOT NULL,
    last_name   VARCHAR(16)     NOT NULL,
```

DEBUG CONSOLE | PROBLEMS | TERMINAL | PORTS

TERMINAL

```
Query OK, 25224 rows affected (0.22 sec)
Records: 25224  Duplicates: 0  Warnings: 0

Query OK, 24919 rows affected (0.23 sec)
Records: 24919  Duplicates: 0  Warnings: 0

Query OK, 24920 rows affected (0.20 sec)
Records: 24920  Duplicates: 0  Warnings: 0

Query OK, 24921 rows affected (0.15 sec)
Records: 24921  Duplicates: 0  Warnings: 0

Query OK, 24922 rows affected (0.17 sec)
```

WE WILLL GET A WARNING AS
I AM USING "select "exercise 1 " as " ";" like this statements to differentiate the exercise output ignore the warning
as
WITH **must be the first statement of a query block**

You cannot place SELECT before WITH
But I am using **before** the WITH clause **only to visually separate outputs**

Because some of the outputs have 10000s of rows of output so I using like this to differentiate differentiate the

```
use employees;
/* 1.Highest salaried employee of Finance */
select "exercise 1 " as " ";
with cte1 as(
select e.emp_no
from employees e
```

```
join salaries s on e.emp_no=s.emp_no
join dept_emp de on e.emp_no=de.emp_no
join departments d on de.dept_no=d.dept_no
where s.to_date='9999-01-01'
and de.to_date='9999-01-01'
and d.dept_name='Finance'
order by s.salary desc
limit 1
)
select emp_no as emp_number from cte1;
```

```
Database changed
+-------------+
|             |
+-------------+
| exercise 1  |
+-------------+
1 row in set, 1 warning (0.00 sec)

+------------+
| emp_number |
+------------+
|      28124 |
+------------+
1 row in set (0.07 sec)
```

```
/* 2.Current manager of Production */
select "exercise 2 " as " ";
with cte2 as(
select dm.emp_no
from dept_manager dm
join departments d on dm.dept_no=d.dept_no
where d.dept_name='Production'
and dm.to_date='9999-01-01'
)
select emp_no as emp_number from cte2;
```

```
+-------------+
|             |
+-------------+
| exercise 2  |
+-------------+
1 row in set, 1 warning (0.00 sec)

+------------+
| emp_number |
+------------+
|     110420 |
+------------+
1 row in set (0.00 sec)
```

```
/* 3.How many managers worked in Human Resources */
select "exercise 3 " as " ";
with cte3 as(
select distinct dm.emp_no
from dept_manager dm
join departments d on dm.dept_no=d.dept_no
where d.dept_name='Human Resources'
)
select count(emp_no) as manager_count from cte3;
```

```
+-------------+
|             |
+-------------+
| exercise 3  |
+-------------+
1 row in set, 1 warning (0.00 sec)

+---------------+
| manager_count |
+---------------+
|             2 |
+---------------+
1 row in set (0.00 sec)
```

```
/* 4.Salary of current manager of Production */
select "exercise 4 " as " ";
with cte4 as(
select s.salary
from dept_manager dm
join salaries s on dm.emp_no=s.emp_no
join departments d on dm.dept_no=d.dept_no
where dm.to_date='9999-01-01'
and s.to_date='9999-01-01'
and d.dept_name='Production'
)
select salary as manager_salary from cte4;
```

```
+-------------+
|             |
+-------------+
| exercise 4  |
+-------------+
1 row in set, 1 warning (0.01 sec)

+----------------+
| manager_salary |
+----------------+
|          56654 |
+----------------+
1 row in set (0.00 sec)
```

```
/* 5.Name starts with a ends with s and 5 letters with salary */
select "exercise 5 " as "";
with cte5 as(
select s.salary
from employees e
join salaries s on e.emp_no=s.emp_no
where s.to_date='9999-01-01'
and e.first_name like 'a___s'
)
select salary from cte5;
```

```
+-------------+
|             |
+-------------+
| exercise 5  |
+-------------+
1 row in set (0.00 sec)

+--------+
| salary |
+--------+
|  50318 |
|  53157 |
|  57924 |
|  95058 |
|  44773 |
|  71353 |
|  56868 |
|  77098 |
| 113980 |
|  81017 |
|  54715 |
|  59754 |
|  66053 |
|  61157 |
|  45739 |
|  50854 |
|  47344 |
|  81327 |
|  73503 |
|  58754 |
|  71376 |
```

```
|  75130 |
|  60143 |
|  54043 |
|  83841 |
|  77002 |
|  48275 |
+--------+
201 rows in set (0.04 sec)
```

Remaining output if needed is in Gurivi_Reddy_Yasaswini_Cte_and_Join_output.txt file

```
/* 6.Hire date and from date difference */
select "exercise 6 " as "";
with cte6 as(
select datediff(de.from_date,e.hire_date) diff
from employees e
join dept_emp de on e.emp_no=de.emp_no
where e.hire_date<>de.from_date
)
select diff as date_difference from cte6;
```

```
+-------------+
|             |
+-------------+
| exercise 6  |
+-------------+
1 row in set (0.00 sec)

+-----------------+
| date_difference |
+-----------------+
|            3908 |
|            3384 |
|             429 |
|            1273 |
|            2649 |
|            3959 |
|            2485 |
|            1906 |
|            1111 |
|            1944 |
|            2530 |
|            1473 |
|            3571 |
|             391 |
|            2095 |
|            2128 |
|            4978 |
|            2394 |
|             116 |
|            3264 |
|            2632 |
```

```
|            2943 |
|            1035 |
|            1602 |
|            2055 |
|             496 |
|              75 |
+-----------------+
181302 rows in set (0.42 sec)
```

Remaining output if needed is in Gurivi_Reddy_Yasaswini_Cte_and_Join_output.txt file

```
/* 7.Highest salaried manager */
select "exercise 7 " as "";
with cte7 as(
select dm.emp_no
from dept_manager dm
join salaries s on dm.emp_no=s.emp_no
where dm.to_date='9999-01-01'
and s.to_date='9999-01-01'
order by s.salary desc
limit 1
)
select emp_no as emp_number from cte7;
```

```
+-------------+
|             |
+-------------+
| exercise 7  |
+-------------+
1 row in set (0.00 sec)

+------------+
| emp_number |
+------------+
|     110039 |
+------------+
1 row in set (0.00 sec)
```

```
/* 8.Manager earning less than highest employee of department */
select "exercise 8 " as "";
with mgr as(
select dm.emp_no,dm.dept_no,s.salary
from dept_manager dm
join salaries s on dm.emp_no=s.emp_no
where dm.to_date='9999-01-01'
and s.to_date='9999-01-01'
),
emp as(
select de.dept_no,max(s.salary) max_salary
from dept_emp de
join salaries s on de.emp_no=s.emp_no
where de.to_date='9999-01-01'
and s.to_date='9999-01-01'
group by de.dept_no
)
select mgr.emp_no as emp_number
from mgr
join emp on mgr.dept_no=emp.dept_no
where mgr.salary<emp.max_salary;
```

```
+-------------+
|             |
+-------------+
| exercise 8  |
+-------------+
1 row in set (0.00 sec)

+------------+
| emp_number |
+------------+
|     110039 |
|     110114 |
|     110228 |
|     110420 |
|     110567 |
|     110854 |
|     111133 |
|     111534 |
|     111939 |
+------------+
9 rows in set (0.45 sec)
```

```
/* 9.Manager with highest salary among all departments */
select "exercise 9 " as "";
with cte9 as(
select dm.emp_no
from dept_manager dm
join salaries s on dm.emp_no=s.emp_no
where dm.to_date='9999-01-01'
and s.to_date='9999-01-01'
order by s.salary desc
limit 1
)
```

```
select emp_no as emp_number from cte9;
```

```
+-------------+
|             |
+-------------+
| exercise 9  |
+-------------+
1 row in set (0.00 sec)
```

```
+------------+
| emp_number |
+------------+
|     110039 |
+------------+
1 row in set (0.00 sec)
```

```
/* 10.Employees not assigned to any department currently */
select exercise10 as "";
with cte10 as(
select emp_no
from dept_emp
where to_date='9999-01-01'
)
select e.emp_no as emp_number
from employees e
left join cte10 c on e.emp_no=c.emp_no
where c.emp_no is null;
```

```
+------------+
| emp_number |
+------------+
|     110039 |
+------------+
1 row in set (0.00 sec)
```

```
+------------+
| emp_number |
+------------+
|      10008 |
|      10011 |
|      10015 |
|      10021 |
|      10025 |
|      10028 |
|      10033 |
|      10034 |
|      10042 |
|      10048 |
|      10055 |
|      10064 |
|      10073 |
|      10075 |
|      10082 |
|      10087 |
|      10090 |
|      10094 |
|      10098 |
```

```
|     499950 |
|     499957 |
|     499958 |
|     499967 |
|     499971 |
|     499975 |
|     499978 |
|     499982 |
|     499988 |
|     499989 |
|     499994 |
+------------+
59900 rows in set (0.42 sec)

mysql> notee
```

Remaining output if needed is in Gurivi_Reddy_Yasaswini_Cte_and_Join_output.txt file